

Exact Diagonal Completion on Reachable Subspaces: Application to QAOA Placement

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Unused encoding states offer opportunities to simplify quantum circuits. For algorithms restricted to reachable subspaces, unspecified diagonal-operator entries can be optimized without changing ideal computation. We investigate exact diagonal completion for the quantum approximate optimization algorithm (QAOA) applied to placement, using permutation-preserving register swaps. We construct exact Manhattan-distance operators through weighted- ℓ_1 optimization of Walsh coefficients and a sparse recurrence requiring $O(\sqrt{m})$ terms on balanced rectangles, avoiding $O(m^4)$ dense constraint storage.

Across 160 geometries, weighted- ℓ_1 completion reduces controlled-NOT (CX) counts relative to four alternative extensions in all 96 cases with unused binary codes under Gray-code synthesis. On an independently specified 60-case cohort, median reductions relative to virtual-coordinate extension are 28.0%, 53.9%, and 21.6% at six, nine, and twelve sites. Under generic diagonal synthesis, reductions decrease to 10.9%, 1.3%, and 0.7%, demonstrating compiler dependence. Additional ancillas reduce mixer serialization, but token circuits remain deeper than one-hot baselines. Ideal placement simulations show baseline-dependent solution quality, with classical search performing better. OpenROAD integration takes 72 QAOA and 216 classical placements across six RTL designs through clock-tree synthesis and global routing with zero overflow. Completion improves phase construction; no end-to-end advantage is established.

CCS Concepts: • **Theory of computation** → **Quantum computation theory**; • **Hardware** → *Physical design (EDA)*; • **Software and its engineering** → *Compilers*.

Additional Key Words and Phrases: constrained QAOA, placement, exact diagonal synthesis, Walsh expansion, XY mixers, partially specified operators

1 Introduction

A quantum circuit is often synthesized from an operator specified on the full Hilbert space, even when the surrounding algorithm can reach only a small part of it. For a diagonal cost operator, entries outside the reachable computational-basis states have no effect on the ideal algorithm. Choosing these entries is a compilation opportunity: different exact extensions of the same required action can have different Walsh spectra and different circuit costs. We call this choice *exact diagonal completion*.

We study completion through a placement problem: assigning circuit cells to physical sites while reducing the lengths of their interconnections (nets). Each cell and each labeled vacancy has a binary register that stores a site index. Partial swaps of entire registers preserve a permutation of the valid site labels, so every measured placement is legal. When the number of sites is not a power of two, some binary labels are unused. A distance operator may take

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arbitrary values on those labels while remaining exact on every state the algorithm visits. The resulting freedom is separate from the familiar task of optimizing a parity network for an already specified diagonal [2, 32, 37].

The contributions are as follows.

- (1) We construct exact Manhattan-distance phases using a weighted- ℓ_1 linear program and a sparse coordinate recurrence. The former handles arbitrary assignments of labels to coordinates; the latter avoids the dense constraint matrix when register bits encode coordinates. We prove their agreement with the required action on the reachable states.
- (2) Across 160 geometries, we show that completion’s gate savings depend strongly on synthesis. L1 completion uses fewer CX gates than four inexpensive extensions in all 96 cases with unused labels under Gray synthesis, but the gain is much smaller under generic diagonal synthesis. A smaller coefficient norm does not certify a minimum-gate circuit.
- (3) We evaluate the consequences for full-circuit resources and placement. Mixer depth and baseline choice limit the benefit, while a separate study on six RTL designs demonstrates that selected legal placements can pass through clock-tree synthesis and global routing. These results establish an integration path without demonstrating a quantum placement advantage.

Section 3 defines the reachable subspaces and circuits. Section 4 develops the completion constructions, followed by the experimental design in Section 5. We present synthesis results first, then full-circuit resources and placement quality in Sections 6–8, including the OpenROAD integration results in Section 8.4. Appendix F collects the principal notation.

2 Related Work

QAOA alternates cost and mixing operations [12]; the quantum alternating operator ansatz extends this approach to constrained spaces, including permutations [18]. XY mixers preserve excitation number [36], while other constructions address assignment, scheduling, general mixer synthesis and Grover mixing [5, 15, 28, 34]. Constraint-preserving annealing drivers provide related antecedents [19]. We use established mixer principles to obtain the placement invariant; the permutation and ring mixers are not new families.

Encoding choices must account for both objective and mixer circuits. Sawaya et al. compare compact, one-hot, Gray and other encodings on this basis [29]. Compact permutation and domain-wall constructions offer additional width–circuit tradeoffs [7, 17], while Prog-QAOA compiles classical objective and constraint programs [3]. Application studies also show sensitivity to penalty and compilation choices [30]. Our full-circuit comparisons follow this perspective and examine the particular costs of a completed distance phase.

Walsh expansions and Gray-code traversal are established methods for diagonal synthesis [37]. Phase-polynomial optimization targets parity-network complexity [2, 27], and phase gadgets and ZX rewriting provide further transformations [10, 23]. We compare with generic diagonal synthesis as implemented in Qiskit [20, 31]. Our related studies optimize routed cost by learning parity-term orders [14] and selecting among orders with equal logical support cost [13]. These methods optimize a specified operator. Completion instead chooses its unspecified entries before applying synthesis, using the freedom studied for partially specified quantum operators [32] and, classically, for logic functions with don’t-care inputs [6]. Our contribution is the distance-specific constructions and their evaluation against alternative extensions and synthesis methods. Neither weighted basis pursuit [8] nor approximate Walsh truncation is introduced here.

The analogy with classical don't-care inputs concerns where correctness is required. In logic minimization, outputs on excluded inputs may be chosen to simplify an implementation; here, diagonal entries on unreachable basis states may be chosen to simplify a phase circuit. The quantum construction must preserve relative phases on every reachable superposition, not merely measurement probabilities for a selected input. Proposition 3 provides this guarantee for any feasible completion, while the resulting cost still depends on the synthesis procedure.

For variational optimization, conditional value-at-risk (CVaR) emphasizes low-cost outcomes [4], and initialization and reachability can affect shallow-circuit performance [1, 11]. Quantum placement and quadratic-assignment studies consider related but distinct subproblems [16, 22, 35]. The small pairwise windows studied here are far simpler than the objectives and scales addressed by classical physical-design tools [9, 21, 25, 26].

3 Placement Encodings and Reachable Subspaces

Let n named cells occupy distinct sites in a set of size $m \geq n$. Site s has coordinate $r_s = (x_s, y_s)$, and a legal assignment is an injection $f : \{0, \dots, n-1\} \rightarrow \{0, \dots, m-1\}$. For weighted pairwise nets E ,

$$C(f) = \sum_{(u,v,w) \in E} w d(f(u), f(v)), \quad d(a, b) = |x_a - x_b| + |y_a - y_b|. \quad (1)$$

For a two-terminal net, this distance equals half-perimeter wirelength (HPWL). Equation (1) is a pairwise approximation to placement cost; it does not model general multi-terminal nets. The OpenROAD integration study uses an anchored local-window objective (Appendix E). There are $P(m, n) = m!/(m-n)!$ feasible real-cell assignments. A circuit with p mixing layers alternates mixing with phases generated by the placement cost. We compare two encodings and their associated mixers.

3.1 One-hot encoding with XY mixing

The one-hot variable $x_{u,s}$ records whether cell u occupies site s . The quadratic unconstrained binary optimization (QUBO) objective is

$$H_{\text{pen}} = \sum_{(u,v,w) \in E} \sum_{s,t} w d(s, t) x_{u,s} x_{v,t} + \lambda \sum_u \left(\sum_s x_{u,s} - 1 \right)^2 + 2\lambda \sum_s \sum_{u < v} x_{u,s} x_{v,s}. \quad (2)$$

The first penalty enforces one site per cell; the second penalizes cells sharing a site. We use collision coefficient 2λ throughout. Preparation sets one bit in every row. Within each row, the mixer applies $R_{XX}(2\beta_t)R_{YY}(2\beta_t)$ in lexicographic site-pair order, using a shared angle β_t . The product equals $\exp[-i\beta_t(XX + YY)]$ and preserves row weight. These operations do not preserve site occupancy: two different rows can choose the same site. Complete per-row connectivity costs $\Theta(pnm^2)$ two-qubit rotations before decomposition into native gates, rather than the linear count of a nearest-neighbor XY chain. The ring control instead uses $(0, 1), (1, 2), \dots, (m-2, m-1), (m-1, 0)$ in that order, once per row and layer, requiring $\Theta(pnm)$ rotations. We refer to these baselines as complete-XY and ring-XY, or collectively as Row-XY because mixing acts within each cell row. Both use nm data qubits and have an invariant basis of size m^n . They are ordered products, not exact evolution under a summed XY Hamiltonian.

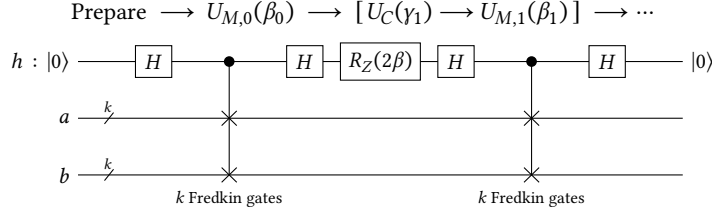


Fig. 1. Prepared token circuit and gate-level expansion of its mixer block. Each controlled register swap is k Fredkin gates sharing the indicated ancilla. The first cost phase is absent because preparation is a basis state.

3.2 Token permutation encoding

Let $k = \lceil \log_2 m \rceil$. Associate a k -qubit register with each of the n real cells and $m - n$ distinguishable vacancy tokens, denoted EMPTY. Initialize real-cell registers to $f(u)$ and assign unoccupied sites, in increasing order, to the EMPTY registers in increasing token order. This requires only basis-state X gates. All m site labels then occur exactly once.

For an edge (a, b) in a connected graph on the token registers, apply

$$U_{ab}(\beta) = \exp(-i\beta S_{ab}) = \cos \beta I - i \sin \beta S_{ab}, \quad (3)$$

where S_{ab} swaps the complete registers. An ancilla initialized to $|0\rangle$ records the ± 1 eigenspace of SWAP, receives the corresponding phase, and returns to $|0\rangle$. The ancilla is clean: it begins and ends in $|0\rangle$. Figure 1 shows the circuit. The serial implementation reuses one; the parallel implementation reserves $\lfloor m/2 \rfloor$ (Appendix B). The serial width is $mk + 1$; for four cells and six sites it is 19 qubits, compared with 24 for Row-XY. An information-theoretic encoding of the real assignments needs only $\lceil \log_2 P(m, n) \rceil$ qubits, without providing an efficient mixer. The serial token circuit uses fewer qubits than one-hot only when $mk + 1 < nm$. For example, at four cells and sixteen sites it uses 65 qubits versus 64 for one-hot, so increasing the number of vacancies can eliminate the width benefit.

PROPOSITION 1 (INJECTIVITY AND MEASUREMENT SEMANTICS). *Starting from a permutation of the m valid site labels, any product of the partial swaps in Eq. (3) and diagonal cost phases remains in the span of valid permutations. Measurement of the real-cell registers therefore always gives an injective placement. Every real assignment has $(m - n)!$ distinct EMPTY-token completions, and its probability is the sum of their basis probabilities.*

PROOF. A register transposition maps a permutation to a permutation, so its linear combination with the identity preserves that span. A diagonal phase cannot change its support. Induction gives the invariant. The unused site labels can be assigned bijectively to the labeled EMPTY registers in $(m - n)!$ ways. Those completions are orthogonal measurement outcomes, so their probabilities, not amplitudes, sum when EMPTY labels are discarded. $\square$

Connected transpositions generate all token permutations, but this statement about reachability does not imply that a shallow ordered circuit distributes amplitude uniformly. The reduced simulator retains all $m!$ token states; it does not merge states that differ only in EMPTY-token labels. A token-label graph can treat different EMPTY tokens differently, so label invariance is not assumed. Distinguishable EMPTY tokens simplify the invariant and distance phase at the cost of extra registers and an enlarged simulation space.

3.3 Ordering is part of the ansatz

Our primary token graph is the line $[(0, 1), (1, 2), \dots, (m-2, m-1)]$. The schedule selected during development applies swaps involving an EMPTY register first, with lexicographic edge order within each group, and uses that same list in every layer. An edge is EMPTY-incident when at least one register index is at least n . The historical routing control uses rotating order: layer ℓ cyclically shifts the fixed list left by $\ell \bmod (m-1)$. The reversed schedule reverses the original fixed list in every layer. Every edge receives the full shared layer angle β_ℓ . Neighboring register swaps generally do not commute, so schedules sharing a transition graph need not share a shallow variational family.

3.4 Removing redundant cost operations

PROPOSITION 2 (REDUNDANT COST OPERATIONS). *For a computational-basis initialization and diagonal cost, the first cost phase changes only global phase. For Row-XY initialized with one excitation per row, the row penalty in Eq. (2) vanishes on every reachable state and can be removed from all phases without changing ideal output probabilities. Collision penalties cannot be removed on this basis.*

PROOF. A basis state $|f_0\rangle$ is an eigenvector of diagonal H , hence $e^{-iy_0 H}|f_0\rangle = e^{-iy_0 H(f_0)}|f_0\rangle$. Every row-XY factor preserves its row's excitation number, and diagonal phases preserve basis support. Thus $\sum_s x_{u,s} - 1 = 0$ on the entire reachable span. The tensor product of the row-weight-one spaces still contains collisions. $\square$

Both prepared circuits consequently use $2p - 1$ free parameters,

$$\theta = (\beta_0, \gamma_1, \beta_1, \dots, \gamma_{p-1}, \beta_{p-1}), \quad |\psi(\theta)\rangle = U_{M,p-1} U_C(\gamma_{p-1}) \cdots U_{M,1} U_C(\gamma_1) U_{M,0} |f_0\rangle. \quad (4)$$

These simplifications preserve the state prepared from $|f_0\rangle$; they need not preserve the unitary on other inputs. Applying them to both encodings ensures that the resource comparison reflects operations needed by the algorithm.

4 Exact Completion of the Distance Phase

Real-cell registers expose both arguments of each net distance directly. For a pair of k -bit registers, let $z = a + 2^k b$ and $W_{zj} = (-1)^{\text{popcount}(z \& j)}$ be the 2^{2k} -dimensional Walsh matrix, where popcount counts set bits and $\&$ denotes bitwise AND. A distance extension has coefficients c_j with $\tilde{d}(z) = \sum_j W_{zj} c_j$. Here j is a bit mask specifying the qubits in the Pauli-Z product Z_j . The corresponding diagonal phase for net weight w is $\exp[-i\gamma w \sum_j c_j Z_j]$. For each nonconstant term, a CX network computes the parity of the selected qubits onto one of them, applies $R_Z(2\gamma w c_j)$, and uncomputes the parity; the constant is recorded as global phase. The phase needs no location-decoding ancilla.

A naive exact extension sets invalid-code distances to zero and transforms that complete table. For $m < 2^k$, this fixes entries the ideal token circuit never queries. Instead define $V = \{a + 2^k b : 0 \leq a, b < m\}$ and solve

$$\min_c \sum_{j>0} [0.01 + 2 \max(\text{popcount}(j) - 1, 0)] |c_j| \quad \text{subject to} \quad W_{V,\cdot} c = d_V. \quad (5)$$

Splitting each coefficient into positive and negative parts yields a linear program (LP), specifically weighted basis pursuit [8]. We call the resulting extension L1 completion. The constant is unpenalized. The weight resembles the isolated CX count of a parity term, with a small positive weight for one-body terms. It multiplies coefficient magnitude, however, so it is a sparsity surrogate, not the discrete gate-count objective and not a proof of minimum CX.

PROPOSITION 3 (VALID-DOMAIN PHASE EQUIVALENCE). *If two coefficient vectors satisfy the equalities in Eq. (5), their distance phases agree on all pairs of valid site labels, for every real γ and net weight. Replacing every token cost phase by either extension therefore preserves the prepared state exactly. If $m = 2^k$, the equalities uniquely fix all Walsh coefficients.*

PROOF. Both diagonal eigenvalues equal $d(a, b)$ at every valid pair, so exponentiation preserves their equality. The token invariant keeps every layer inside that domain; induction through the alternating product proves the prepared-state claim. At full binary capacity, V is the complete index set and the Walsh matrix is invertible. $\square$

For computed coefficients with maximum valid-pair residual ϵ , the induced cost error on any legal assignment is at most $\epsilon \sum_{(u,v,w) \in E} |w|$. The corresponding restricted phase-unitary error in operator norm is at most $|\gamma| \epsilon \sum_E |w|$, by $|e^{ia} - e^{ib}| \leq |a - b|$. Thus numerical phase error scales with angle and total net weight. We report the underlying distance residuals.

We deliberately constrain equal-site pairs as well as distinct pairs. They are unnecessary for injective token states but strengthen the distance specification. The freedom resides only in invalid binary labels. It preserves ideal action at layer boundaries; it does not guarantee identical behavior under physical noise during the two different decompositions.

The dense linear program has $2^{2k} = O(m^2)$ coefficients, m^2 equalities, and $O(m^4)$ explicitly stored constraint entries. It is practical at the small sizes studied here, not a scalable oracle-construction theorem. A generic two-register phase has $O(m^2)$ Walsh terms. Independent parity computation costs $O(m^2 \log m)$ CX per net; a full reflected-Gray traversal costs $O(m^2)$ CX per net before inter-net aggregation. Coordinate structure and term sharing can reduce this count substantially. Conversely, the token mixer on a line or ring requires $O(pm \log m)$ register-swap work, so the phase can dominate an otherwise compact mixer. Algorithm 2 includes the full eigenspace-resolution cost in that mixer bound.

4.1 Structured completion for coordinate-indexed arrays

Arbitrary site-label permutations need not preserve geometric sparsity. A special case does admit direct construction: unsigned coordinate bits in row-major order, including a valid prefix of a binary-capacity rectangle.

LEMMA 1 (EXACT COORDINATE DISTANCE AND WALSH SUPPORT). *For ℓ -bit unsigned integers a, b , let a', b' be their lower $\ell - 1$ bits and $A = (-1)^{a_{\ell-1}}, B = (-1)^{b_{\ell-1}}$. Their distance obeys*

$$|a - b| = \frac{1 + AB}{2} |a' - b'| + 2^{\ell-2} (1 - AB) + \frac{B - A}{2} (a' - b'), \quad D_0 = 0, \quad (6)$$

where D_0 is the distance between two zero-bit integers. For $\ell \geq 1$, the Walsh expansion has exactly

$$T_\ell = 5 \cdot 2^\ell - 4\ell - 4 \quad (7)$$

nonzero coefficients, including the constant.

PROOF. If the high bits agree, the first term of Eq. (6) remains and the other two vanish. If they differ, their ordering fixes the sign of $a - b$ and gives the displayed low-bit correction. This proves the distance identity. Expand $a' - b' = \sum_{i < \ell-1} 2^{i-1} (Z_{b_i} - Z_{a_i})$ and accumulate identical masks. The base case is $T_1 = 2$. For $\ell \geq 2$, each old mask appears both alone and multiplied by AB ; the $4(\ell - 1)$ new high-low two-bit masks are distinct from both sets. Only the constant and AB coefficients combine with the middle term of Eq. (6). The new constant is the positive mean distance. For $N = 2^{\ell-1}$, the previous constant is $(N^2 - 1)/(3N)$, so the new AB coefficient is $-(2N^2 + 1)/(6N)$ and is also nonzero. Hence $T_\ell = 2T_{\ell-1} + 4(\ell - 1)$, which gives Eq. (7) by induction. $\square$

This constructs a sparse list of coefficients without a distance table or LP. Construction uses $O(2^\ell)$ arithmetic operations and entries. Exact coefficients (integers divided by powers of two) and masks require $O(\ell)$ bits per entry, adding an $O(\ell)$ factor when accounting for bit operations and storage.

For coordinate widths $k_x + k_y = k$, combine the two coefficient lists on their separate bit supports. The bound is $O(2^{k_x} + 2^{k_y})$ terms and $O(k(2^{k_x} + 2^{k_y}))$ independent-parity CX per net, or $O(k\sqrt{m})$ for balanced binary rectangles. This is a subclass guarantee, not an arithmetic oracle for arbitrary shuffled coordinates. The extension preserves all valid prefix distances; unused grid points are virtual, not newly legal sites. A change in the number of physical sites changes the optimization problem, whereas extending the binary truth table does not.

4.2 From coefficients to circuits

Each nonconstant Walsh term requires a rotation conditioned on a parity. Computing and uncomputing every parity independently can repeat many CX gates. Our primary synthesis method first combines terms from different nets that act on the same global qubit support. It groups the remaining terms by their highest-index qubit, computes parity onto that qubit, and visits the control masks in reflected Gray-code order. Between rotations, only controls that change are updated; each target is restored at the end of its group. We refer to this method as Gray synthesis.

Two additional ordering methods provide comparisons. Fixed mask order processes nets and their integer-valued masks in increasing order. Beam search instead retains four candidate term sequences and extends each with up to ten terms chosen by support overlap, ranking sequences by CX count after adjacent cancellation. These methods retain separate terms for each net. Their comparison with Gray synthesis therefore includes both term aggregation and ordering. Algorithm 3 specifies the ordering and tie rules. All three methods retain nonzero coefficients after the numerical threshold; intentional approximation is evaluated separately.

4.3 Alternative extensions and approximation

We compare five extensions while holding coordinates and site labels fixed. Besides zero and weighted- ℓ_1 , a virtual-coordinate extension appends unused integer lattice points in row-major order inside the bounding rectangle, extending rows above it as necessary. Manhattan distance then specifies every binary-code pair. A nearest-code extension maps each invalid code to a valid code of least Hamming distance, breaking ties by smaller code, and uses the mapped pair's distance. Mean fill assigns the mean of all m^2 valid ordered-pair distances whenever either label is invalid. All five extensions retain every valid pair, including equal-site pairs. Virtual geometry does not imply sparse coefficients when the original labels are shuffled, and zero padding is not proved to maximize support or gate count.

For L1 and virtual completion, the approximate control tests coefficients in increasing absolute magnitude, breaking ties by mask. It removes a term only when the maximum valid-pair residual remains at most ηD , where D is the valid-site diameter and $\eta \in \{.001, .01, .05\}$. We report the achieved residual and the resulting assignment-error bound, not just the allowed tolerance. This greedy truncation is not an optimal sparse approximation. It measures circuit size against bounded objective error without retraining the ansatz.

Symbolic beam and Gray circuits retain a free γ . Numeric comparisons fix $\gamma = .371$ and compare Gray, Qiskit's generic diagonal synthesis and the ZX-calculus optimizer PyZX 0.10.3, using full reduction followed by circuit extraction. Each extracted circuit must pass PyZX's equality check against its parsed input; independent small Qiskit unitary checks test the conversion and all five extension policies. OpenQASM angle conversion is checked separately. Fixing angles can enable additional simplifications, so numerical and symbolic results are reported separately.

5 Experimental Design

The evaluation asks three questions: how completion changes phase synthesis, how those savings affect full-circuit resources, and what the motivating encoding comparison implies for placement quality. Phase circuits isolate the first question. Resource comparisons include preparation, mixing and cost phases. Quality comparisons use ideal simulation and do not measure device performance. A separate OpenROAD study checks whether selected local placements can be used in a physical-design flow (Section 8.4).

5.1 Geometries and placement instances

The phase study uses two cohorts, both with four cells and six weighted nets. The initial cohort contains 100 cases: twenty at each of $m = 6, 8, 9, 12, 16$, spanning four geometry families and five coordinate/label realizations. The replication cohort contains twelve cases at each size, using compact, irregular-line, L-shaped and scatter geometries, shuffled site labels and connected weighted nets. Its cases were fixed before their outcomes were evaluated. Together, the cohorts contain 96 cases with unused binary codes and 64 power-of-two controls, for which no completion freedom exists. Later comparisons of extensions and synthesis methods reuse these same 160 cases; they are robustness tests, not additional replications.

The placement study distinguishes initial confirmation from transfer to a changed generator. Confirmation comprises 36 four-cell/six-site and twelve five-cell/seven-site instances, drawn from sparse and dense grid, line and L-shaped families. One four-cell instance had been used for a smoke test; we report both the full cohort and its exclusion sensitivity. Transfer uses twelve instances at each of four cells/six sites, five cells/eight sites and six cells/eight sites, with compact, irregular-line and scatter geometries. The two cohorts differ in geometry and sampling and are not paired with each other. Methods are paired within an instance.

The primary token schedule and the initial complete-XY penalty $\lambda = 5$ were selected during development. Subsequent controls were specified after those earlier results were known, but before their own execution. They assess robustness to baseline and implementation choices; they do not provide external preregistration or an independent application sample. Appendix A records the sequence and source freezes.

5.2 Circuit synthesis and resource measurement

All full circuits include basis preparation, omit the redundant operations identified in Section 3, and retain $2p - 1$ free symbolic parameters. Measurements are excluded from gate counts. Logical circuits use the rz, sx, x, cx basis with Qiskit optimization level 3 and seed 123. CX denotes a controlled-NOT gate. Routing uses Qiskit 2.5.2 with SABRE [24] layout and routing on three targets: an archived 127-qubit heavy-hex FakeSherbrooke target, a 25-vertex path (line25) and a 5×5 grid (grid25). The native two-qubit gate is echoed cross-resonance (ECR) on Sherbrooke and CX on the synthetic graphs. Counts are compared within each target and are not pooled across gate bases.

The main full-circuit comparison uses the twelve six-site replication phase cases, $p = 3$, L1 completion and Gray synthesis. It includes serial and parallel token circuits and both Row-XY baselines, with five routing seeds per target. Each circuit is identical across targets before routing. For each method and instance we first take the median across seeds, then compute paired percentage changes and summarize their median and interquartile range (IQR). Routing seeds do not increase the number of independent instances. Earlier compiler and synthesis configurations are reported separately in Appendix A.

For parallel mixing, disjoint register swaps are assigned to the earliest stage consistent with all shared-register dependencies. This preserves the ordered unitary. At six sites, the five swaps occupy three stages and use at most two ancillas simultaneously, although three are reserved. Routing includes the reserved wire: the serial and parallel token circuits have 19 and 21 qubits, respectively, compared with 24 for Row-XY. Statevector checks include arbitrary superpositions and verify that every ancilla returns to zero.

5.3 Training and penalty selection

Unless depth is varied explicitly, placement circuits use $p = 3$ and five free parameters. Training minimizes conditional value-at-risk (CVaR), the mean of the lowest-cost quarter of the output distribution, using the derivative-free optimizer COBYLA with a 200-objective-call cap, initial radius 0.2 and tolerance 0.001. Every method uses the same three legal starting placements: deterministic, random and poor. The poor start is an exact-reference worst placement and serves as a stress test. Confirmation uses four parameter seeds; transfer uses two. Four-cell transfer also includes training with 2,048 ideal samples per call, averaging the lowest 512 energies. Its final exact probabilities are evaluation diagnostics and are not fed back to training.

For each Row graph, eighteen disjoint development instances are used to choose one of four penalties: $\lambda = 5, B/4, 1.000001B$ or $4B$. Selection maximizes mean optimum probability within each problem size. Here $B = Dd_{\max}/2$, D is the site-set diameter, and $d_{\max}$ is the maximum weighted net degree. For nonnegative net weights and $m \geq n$, $\lambda > B$ makes every global energy minimizer in the one-hot subspace legal: moving a colliding cell to an empty site increases wirelength by at most $Dd_{\max}$ while removing at least 2λ in penalties. This bound concerns exact energy minimization, not the feasibility of shallow-QAOA samples. Complete and ring penalties are selected independently before transfer evaluation; fixed- λ results are retained alongside calibrated results.

To compare quality at similar circuit cost, we fix serial token at $p = 3$ and route each Row baseline at $p = 1, \dots, 12$. For each target and metric (native two-qubit count or depth), we select the largest Row depth whose cohort median does not exceed token's. Selection uses resource counts alone, with the same twelve transfer instances, five routing seeds and deterministic preparation. It matches cohort medians, not every instance or training start. Selected circuits receive the same 200-call training cap. Consequently, neither per-call work nor total training cost is equalized.

5.4 Quality metrics and statistical analysis

The primary placement metric is the probability of measuring an optimal legal assignment, called optimum probability below. We also report legal probability, optimum probability conditional on legality, and expected best cost from 128 or 512 final measurements. The conditional probability is computed within each run before averaging. Best-of-sample selection retains the known starting placement as a fallback and charges illegal measurements against the sampling budget.

We average starts and optimizer seeds within each instance before pairing methods. We report 95% percentile confidence intervals from 20,000 instance-bootstrap draws stratified by geometry family. The small families and restricted generators limit population-level interpretation. Additional endpoints receive descriptive exact sign and Wilcoxon signed-rank tests, with Holm correction across nine existing conditions or twelve resource-budget comparisons, separately for each endpoint and test. The Wilcoxon test assumes symmetric paired differences; bootstrap intervals are not adjusted for multiplicity. Actual optimizer calls, seeds, shot counts and numerical conventions are retained in the artifact and summarized in Appendix A.

Table 1. Exact extensions under symbolic Gray synthesis. The initial 100-case and replication 60-case cohorts contain 20 and 12 geometries per site count, respectively. CX counts are medians with interquartile ranges. Percentage changes are medians of paired L1/reference changes, not ratios of the displayed medians; negative values favor L1. Power-of-two controls are omitted because all extensions coincide.

Cohort	m	L1 CX [IQR]	vs zero	vs virtual	vs nearest	vs mean
100-case	6	238 [236, 262]	-23.8%	-26.3%	-16.7%	-23.8%
100-case	9	655 [619, 673]	-55.0%	-55.0%	-52.5%	-55.0%
100-case	12	1095 [991, 1164]	-24.7%	-23.0%	-23.7%	-24.7%
60-case	6	228 [204, 235]	-27.6%	-28.0%	-25.2%	-27.6%
60-case	9	552 [438, 630]	-53.9%	-53.9%	-53.0%	-53.9%
60-case	12	939 [858, 966]	-21.3%	-21.6%	-21.5%	-21.3%

The quality analyses contain 18 Holm-corrected families: five endpoints across nine original conditions and four endpoints across twelve resource-budget comparisons, each evaluated with two tests. These families contain 186 individual test results. Correction is kept within each endpoint and test to address distinct questions about legality, solution quality and sampling cost; it does not control the family-wise error rate across the whole paper. The corrected values and unadjusted intervals are therefore evidence about the specified comparisons, rather than a joint claim of superiority across methods and endpoints. Phase and resource counts are descriptive paired comparisons, not additional significance tests.

6 Completion and Synthesis Results

We first evaluate the distance phase independently of the mixer. This isolates the effect of choosing unspecified table entries, then asks how that effect depends on geometry, label structure and synthesis.

6.1 Completion gains and synthesis dependence

We find that completion improves on all four inexpensive extensions under the same Gray synthesis method (Table 1). L1 uses fewer CX gates than each alternative in all 96 cases with unused binary codes. All 64 power-of-two cases tie within every synthesis method, as required by the uniqueness of their Walsh expansion. On the replication cohort, median reductions relative to virtual-coordinate extension are 28.0%, 53.9% and 21.6% at six, nine and twelve sites. The corresponding reductions relative to nearest-code extension are 25.2%, 53.0% and 21.5%. Thus, on these geometries and label assignments, optimizing the extension improves on natural alternatives to zero padding.

The benefit is sensitive to synthesis. With the angle fixed at $\gamma = 0.371$, generic diagonal synthesis reduces the replication-cohort L1/virtual savings to 10.9%, 1.3% and 0.7%. At these three sizes, L1 uses fewer/equal/more CX gates in 11/0/1, 8/4/0 and 11/1/0 cases, respectively. L1 also improves on virtual extension throughout the replication cases with unused codes under the tested PyZX procedure, but that procedure produces more CX gates than numerical Gray synthesis for every input (Appendix D). The large Gray savings therefore describe a particular combination of completion and synthesis, rather than a compiler-independent reduction.

We verify that these gate savings preserve the required action by reconstructing all 1,760 stored coefficient vectors. The maximum valid-distance residual is 5.69×10^{-14} for the exact extensions. Replaying all 864 saved confirmation and transfer parameter vectors with L1-completed token energies gives the same energies and output probabilities at stored floating-point precision. These checks support Proposition 3: the completion changes implementation cost

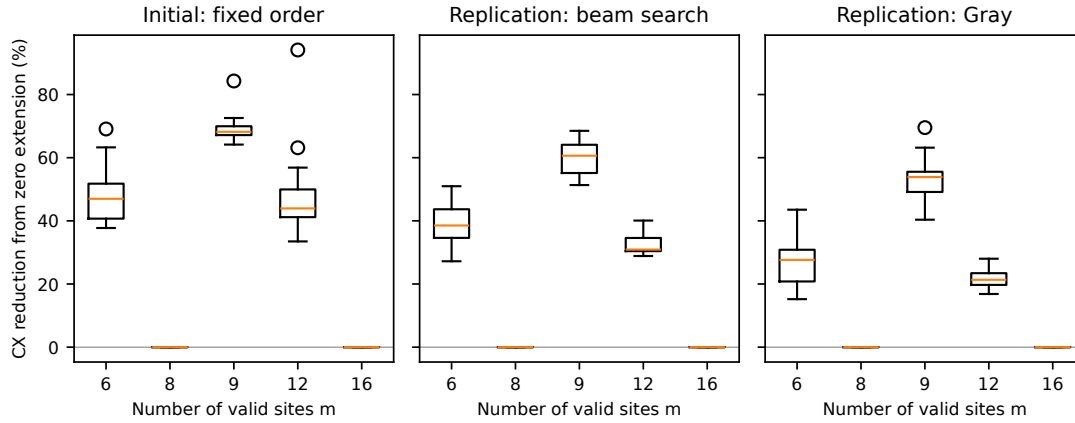


Fig. 2. CX reductions from L1 completion relative to zero extension. The initial cohort uses fixed mask order (20 cases per size); the replication cohort uses beam search and Gray synthesis (12 per size). Boxes show medians and quartiles; whiskers reach the most extreme values within 1.5 IQR, with other observations shown individually. Powers of two have no completion freedom and hence zero reduction.

while preserving the ideal algorithm. Numerical angle conversion and deliberate approximation have separate error checks in Appendix D.

6.2 The extent of completion freedom

A two-register distance table has 2^{2k} entries but only m^2 prescribed values. Its fraction of unspecified entries is therefore

$$F = 1 - \frac{m^2}{2^{2\lceil \log_2 m \rceil}}. \quad (8)$$

This fraction concerns unused binary codes, not the number of vacant placement sites. At a power of two, $F = 0$ and completion cannot change the operator. Between powers of two, freedom becomes available but does not by itself determine the saving: six and twelve sites both have $F = 7/16$, yet their CX reductions differ (Fig. 2). Nine sites have $F = 175/256$. Geometry and label assignment determine the required values, while net aggregation and parity synthesis determine how a chosen extension translates into gates.

6.3 Joint optimization of coordinate contributions

Manhattan distance is additive in its coordinates, suggesting a simpler alternative to joint completion: solve Eq. (5) separately for $|x_a - x_b|$ and $|y_a - y_b|$, then add the coefficient vectors. The sum $c_x + c_y$ is feasible for the joint problem. Consequently, the joint optimum cannot have a larger weighted norm, although its synthesized circuit need not have fewer gates. Both formulations retain the same $O(m^4)$ dense constraint storage because both use the full site-label domain. Coordinate separation alone does not solve the storage problem.

Joint completion has a strictly smaller weighted norm in 67 of the 96 cases with unused codes. Coordinate-separated completion increases replication-cohort Gray CX by median 10.0%, 16.3% and 14.6% at six, nine and twelve sites (Table 2). Some individual cases nevertheless favor separation in gate count, illustrating the distinction between the LP

Table 2. Coordinate-separated relative to joint L1 completion. Each row contains 20 initial-cohort or 12 replication-cohort cases. W/T/L counts cases with fewer/equal/more CX gates for the separated construction. Changes and IQRs summarize paired percent-ages. Norm comparisons in the text evaluate the LP objective on $c_x + c_y$. Power-of-two ties are omitted.

Cohort	Synthesis	m	CX change	IQR	W/T/L
100-case	fixed order	6	+6.6%	[+0.0, +26.9]	0/9/11
100-case	fixed order	9	+12.1%	[+0.0, +27.7]	1/6/13
100-case	fixed order	12	+21.6%	[+0.0, +36.1]	0/6/14
60-case	beam search	6	+11.6%	[+0.0, +22.1]	1/3/8
60-case	beam search	9	+17.0%	[+6.6, +27.7]	0/3/9
60-case	beam search	12	+17.1%	[+7.2, +25.7]	0/3/9
60-case	Gray	6	+10.0%	[+0.0, +19.0]	1/3/8
60-case	Gray	9	+16.3%	[+2.1, +28.9]	0/3/9
60-case	Gray	12	+14.6%	[+4.9, +18.8]	0/3/9

Table 3. Gray CX counts for regular-coordinate distance phases with six nets. Each row is one deterministic geometry. All three extensions are exact on valid pairs; the recurrence assigns distances to unused binary coordinates. These are construction examples rather than a sampled cohort.

Coordinate set	m	Zero CX	L1 CX	Structured CX
row	6	292	238	264
row	8	264	264	264
row	9	1406	344	720
row	12	1286	706	720
row	16	720	720	720
grid	6	318	86	86
grid	8	86	86	86
grid	9	1262	110	148
grid	12	1182	110	148
grid	16	148	148	148

objective and the synthesized circuit. All 64 power-of-two cases yield identical coefficients, and the maximum valid-distance residual is 8.35×10^{-14} . Joint optimization can exploit cancellation between coordinate contributions that separate solves miss.

6.4 Direct construction on regular arrays

The recurrence in Section 4.1 takes a different approach: it uses coordinate bits directly and avoids the dense LP. We evaluate it on binary rows and row-major rectangles, including prefixes with unused codes, at the same five site counts. Each comparison uses a four-cell, six-net phase and Gray synthesis (Table 3). These deterministic geometries examine the construction’s mechanism; they are separate from the shuffled-label cohorts.

The recurrence reproduces every coordinate-pair distance through six bits, and its support count agrees with Eq. (7) through sixteen bits. The latter check constructs coefficients rather than a placement circuit at that width. At six grid sites, the recurrence and L1 both use 86 CX, compared with 318 for zero extension. The recurrence is not always as economical as the LP: at nine grid sites it uses 148 CX versus 110, and at nine row sites 720 versus 344. It therefore provides a more compact construction on this structured subclass without necessarily minimizing the resulting circuit.

Table 4. Full-circuit resource comparison: medians and interquartile ranges of paired percentage changes after five-seed instance medians ($N = 12$). All methods use $p = 3$; token uses L1 completion and Gray synthesis. Circuits are identical across targets before routing. Positive changes are more costly. The native two-qubit gate is ECR on Sherbrooke and CX on the synthetic line25/grid25 targets.

Target	Token	Reference	2q (%) [IQR]	Depth (%) [IQR]
Sherbrooke	parallel	serial	-0.8 [-1.7, +0.4]	-20.9 [-21.4, -19.3]
Sherbrooke	parallel	complete	+1.0 [-3.9, +3.3]	+81.1 [+64.9, +103.2]
Sherbrooke	parallel	ring	+22.4 [+17.7, +28.1]	+113.6 [+94.6, +146.7]
Path	parallel	serial	-0.3 [-0.5, +0.1]	-24.3 [-24.7, -22.9]
Path	parallel	complete	+24.2 [+16.8, +28.0]	+79.6 [+63.9, +94.6]
Path	parallel	ring	+43.2 [+36.4, +50.0]	+109.4 [+92.1, +125.9]
Grid	parallel	serial	-0.1 [-0.4, +0.3]	-19.9 [-20.6, -17.8]
Grid	parallel	complete	-11.6 [-16.2, -10.7]	+15.4 [+5.2, +23.5]
Grid	parallel	ring	+13.3 [+9.2, +15.5]	+44.1 [+39.9, +53.5]

Table 5. Standalone component accounting on the same 12 cases: median logical counts in the rz, sx, x, cx basis. Preparation initializes the deterministic placement; mixer is the first ordered layer; phase is one distance/penalty layer. Data and Anc. are reserved logical widths, including qubits idle in a standalone block; ancillas are clean workspace. R_Z is a subset of 1q gates. Standalone counts are not additive estimates of an optimized full circuit; cross-component cancellation and routing change the total. Routed component medians and IQRs for all three targets are in the artifact.

Method	Component	Data	Anc.	1q	R_Z	CX	Depth
serial	preparation	18	1	7	0	0	1
serial	mixer	18	1	384	282	195	276
serial	phase	18	1	128	128	228	302
parallel	preparation	18	3	7	0	0	1
parallel	mixer	18	3	389	285	195	170
parallel	phase	18	3	128	128	228	302
complete	preparation	24	0	4	0	0	1
complete	mixer	24	0	1080	636	240	147
complete	phase	24	0	210	210	372	106
ring	preparation	24	0	4	0	0	1
ring	mixer	24	0	540	312	96	106
ring	phase	24	0	210	210	372	106

7 Full-Circuit Resources

Phase savings are only one part of the encoding comparison. Token circuits use fewer data qubits, but their swaps and parity networks introduce depth that one-hot circuits may avoid. We compare complete circuits on the same twelve six-site phase instances with five SABRE routing seeds per target [24]. Every token circuit uses L1 completion and Gray synthesis; all methods use $p = 3$ and symbolic angles.

Additional ancillas reduce token depth by allowing disjoint swaps to run concurrently (Table 4). Relative to the serial implementation, the parallel circuit reduces median depth by 20.9% on Sherbrooke, 24.3% on the path and 19.9% on the grid, with changes below 1% in median native two-qubit count. The Sherbrooke depth reduction has a paired IQR of 19.3–21.4%. The remaining comparison depends strongly on the one-hot mixer. On Sherbrooke, parallel token is close to complete-XY in ECR count (+1.0%), but has 81.1% more depth. Against ring-XY, it has 22.4% more ECR gates and 113.6% more depth. On the grid, token uses 11.6% fewer CX gates than complete-XY but 13.3% more than ring-XY; its depth is greater than both. Reducing serialization helps, but does not establish an overall resource advantage.

Table 5 explains why gate and depth rankings differ. The completed token phase has median 228 logical CX gates versus 372 for either Row phase, yet its depth is 302 versus 106. Parallel workspace lowers the token mixer depth from 276 to 170 while leaving phase depth unchanged. The complete and ring mixers use 240 and 96 CX gates, respectively, compared with token’s 195. A parity construction can therefore save gates while exposing less parallelism. These standalone blocks locate the costs; their depths cannot be added to predict an optimized routed circuit because cancellation and scheduling also occur across block boundaries.

8 Placement Quality

Completion preserves token’s ideal output distribution, so its circuit savings do not require a new quality comparison. The experiments here instead assess the token encoding relative to the one-hot baselines that motivate it. They show a favorable four-cell result under some conditions, but no consistent advantage across baselines, sizes and sampling metrics.

8.1 Legality and solution quality

At matched $p = 3$, calibrated four-cell transfer with exact-distribution training gives mean optimum probabilities of 0.05713 for token, 0.01304 for complete-XY and 0.01416 for ring-XY. Ring closes only 2.5% of the positive token-complete difference. Part of the difference reflects legal probability: token is always legal, while the corresponding Row means are 0.396 and 0.412. Conditioning each run on legal outputs gives optimum probabilities of 0.05713, 0.02747 and 0.02195. The token-ring conditional difference has a positive unadjusted bootstrap interval, but the Wilcoxon comparison does not survive Holm correction ($p = 0.448$). These outcomes reflect the combined encoding, mixer order, initialization and optimizer; they do not isolate the causal effect of legality.

Baseline selection also matters within this four-cell setting. As a sensitivity analysis, an oracle chooses the better Row graph and penalty for each evaluation instance and metric. This oracle is not an available tuning procedure because it uses evaluation outcomes. Against it, token’s conditional-probability difference is 0.00602 with interval $[-0.02239, 0.03656]$, and its expected best cost after 512 measurements is higher by 0.606, with interval $[0.083, 1.128]$. Detailed conditional and sampling comparisons appear in Appendix C.

The quality difference also weakens under transfer and sampled training. With 2,048 samples per training call, four-cell token optimum probability falls to 0.02637, compared with 0.01559 and 0.01533 for calibrated complete and ring mixers. Its paired interval against complete-XY includes zero. At five cells/eight sites, token’s mean of 0.000447 is below both fixed and calibrated complete-XY (0.000703 and 0.000821). At six cells/eight sites, token trails fixed complete-XY and exceeds the calibrated mean, but wins only four of twelve paired instances. Intervals against calibrated complete-XY include zero at both larger sizes.

8.2 Quality at comparable circuit cost

Matching layer count does not match circuit cost. Under the cohort-median budgets in Table 6, Row circuits can use two to six layers for the cost of token at $p = 3$; none reaches the $p = 12$ search limit. On Sherbrooke and the path, the depth budget admits complete-XY at $p = 5$ and ring-XY at $p = 6$. Their mean optimum probabilities are 0.00916 and 0.02698, compared with token’s 0.05713. All twelve selected-budget optimum-probability differences have positive unadjusted bootstrap intervals. However, several budgets select the same trained circuit, and their repeated entries are not independent observations.

Table 6. Quality at measured cohort-median budgets on twelve four-cell/six-site transfer instances. Used is the selected Row cost divided by token’s cost, using the cohort median of five-seed instance medians. Quality averages three starts and two optimizer seeds within instance. Positive optimum-probability differences favor token; intervals are paired, family-stratified bootstrap 95% intervals. The artifact includes conditional quality, best-of-128 cost, legal probability and Holm-corrected tests for all twelve resource selections.

Target	Row graph	Budget	p	Used (%)	Token–Row optimum [95% CI]
Sherbrooke	complete	2q	3	98.6	+0.046 [+0.025, +0.067]
Sherbrooke	complete	depth	5	88.9	+0.048 [+0.030, +0.066]
Path	complete	2q	3	84.1	+0.046 [+0.025, +0.067]
Path	complete	depth	5	83.3	+0.048 [+0.030, +0.066]
Grid	complete	2q	2	62.5	+0.038 [+0.015, +0.062]
Grid	complete	depth	3	73.2	+0.046 [+0.025, +0.067]
Sherbrooke	ring	2q	3	81.2	+0.038 [+0.007, +0.065]
Sherbrooke	ring	depth	6	93.5	+0.030 [+0.010, +0.048]
Path	ring	2q	3	71.5	+0.038 [+0.007, +0.065]
Path	ring	depth	6	88.9	+0.030 [+0.010, +0.048]
Grid	ring	2q	3	87.9	+0.038 [+0.007, +0.065]
Grid	ring	depth	4	84.0	+0.038 [+0.017, +0.056]

The inference remains sensitive to the endpoint and multiplicity correction. Against depth-matched ring at $p = 6$, the conditional difference is 0.01275 with interval $[-0.01553, 0.04020]$. Token’s best-of-128 cost is lower by 0.439, with difference interval $[-0.931, -0.018]$, but the Holm-adjusted Wilcoxon p -value is 0.879; for unconditional optimum probability it is 0.212. The positive unadjusted intervals consequently do not establish superiority across the twelve budget comparisons. These comparisons also hold the penalty and 200-call training cap fixed, rather than fully optimizing each circuit family.

8.3 Classical reference searches

Uniform feasible sampling, simulated annealing and multistart greedy search count every objective query, including initialization and repeated proposals. At 512 queries, multistart greedy’s mean best-cost/exact-optimum ratios are 1.0000, 1.0055 and 1.0063 at four, five and six cells. The corresponding ratios from 512 final measurements of analytically trained token circuits are 1.0633, 1.3375 and 1.1729. Quantum training adds work; sampled training alone can consume 409,600 measurement outcomes. Thus the classical methods obtain better solutions even before charging QAOA for training. These small instances support the circuit-construction study, but do not demonstrate a competitive quantum placement solver.

8.4 Integration with a physical-design flow

The placement method also produces assignments that can be used in a conventional physical-design flow. In a separate study of 72 four-cell, six-site windows from six RTL designs, all 72 selected QAOA assignments and 216 classical assignments complete OpenROAD clock-tree synthesis and global routing with zero overflow (Table 7). This is a concrete integration result for legal local placements; it does not measure the effect of diagonal completion on downstream design quality. The search comparison remains favorable to classical methods, including on the 48 windows that are not initially optimal. This study uses a different objective and token schedule from the synthetic experiments; Appendix E describes the design selection, sampling and downstream checks.

Table 7. OpenROAD integration and local-search outcomes on six RTL designs. Optimum counts refer to windows: QAOA uses 4,096 final measurements per optimizer seed, with four seeds and additional training; each of the two recorded budgeted classical searches solves all windows by 360 objective queries per seed. These are not equal-work comparisons. The downstream row counts selected assignments, with three classical methods per window; downstream success means clock-tree synthesis and global routing with zero overflow. The 72 windows are nested within six designs.

Outcome	QAOA	Classical
Optimum found, all windows	71/72	72/72
Optimum found, initially suboptimal windows	47/48	48/48
Downstream flow completed, assignments	72/72	216/216

9 Discussion and Limitations

We show that an invariant offers more than a way to enforce constraints: it also creates freedom to simplify the operator implementing the objective. In this application, choosing entries outside the reachable subspace through weighted- ℓ_1 completion improves on inexpensive distance extensions under Gray synthesis. The much smaller improvement under generic diagonal synthesis shows why completion cannot be assessed in isolation from the compiler. Similarly, the full-circuit results show why a phase saving alone is insufficient: mixer cost, available parallelism and hardware connectivity can determine the final resource ranking.

The two constructions address different representation choices. Joint and coordinate-separated LPs both require $O(m^4)$ dense storage. The recurrence avoids this cost when register bits have the required coordinate meaning; arbitrary label permutations need not preserve its sparse support. Our comparisons hold labels fixed within each case. Optimizing the label assignment itself is a further compilation choice, and coordinate encodings for irregular or sparse site sets may require a different width. Arithmetic or reversible-lookup distance circuits, domain-wall encodings and additional completion of the Row operator remain outside the comparison.

The token encoding is most attractive when the number of vacancies is small. Distinguishable EMPTY registers introduce $(m - n)!$ redundant labels and can erase the width benefit as vacancies increase. The simulations retain all $m!$ token states and cover only four to six cells. They identify neither a scaling crossover nor a noisy-device benefit. Finite-shot training is limited to four cells; the routing targets omit noise, drift, crosstalk and readout effects. Extra ancillas may incur physical costs absent from gate counts. Section 8.4 establishes a practical integration path for the placements, while its classical comparisons reinforce the absence of a solver advantage at this scale. A stronger application claim would require gains that survive mixer and routing costs, training and readout budgets, and comparison with classical search on larger instances; the present results do not locate such a regime.

10 Conclusion

Exact diagonal completion uses freedom outside the reachable subspace to change circuit cost without changing the ideal algorithm. For Manhattan placement phases, weighted- ℓ_1 completion reduces Gray-synthesized CX counts relative to four inexpensive extensions on all 96 evaluated cases with unused codes. A sparse coordinate recurrence provides a direct construction for regular arrays and avoids the dense LP, although it does not always match the LP's gate count. The size of the benefit depends on synthesis, and full-circuit depth remains sensitive to the mixer and connectivity. Placement quality is likewise baseline-dependent, with classical search stronger on the tested instances. Our results make the choice of an exact extension a concrete circuit-design decision: preserve the required action, choose the remaining entries together with the synthesis method, and measure the benefit in the complete implementation.

11 Artifact Availability

The reproducibility artifact is available at [doi:10.5281/zenodo.22970077](https://doi.org/10.5281/zenodo.22970077). It contains the source code, study protocols, benchmark instances, numerical results, coefficient vectors, analysis scripts, dependency specifications and manuscript sources, together with instructions for verification and reproduction. Code is released under the MIT license; original data and documentation are released under CC BY 4.0. Third-party material retains its existing license.

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A Study Provenance and Additional Settings

The study began with small placement and mixer experiments used for development. Their overlapping leave-one-out folds are not independent confirmation. The initial confirmation protocol froze the selected token schedule, penalty, instances and endpoints before the full run. One case, `fresh4_sparse_grid_904000`, already had sixteen smoke-test runs; it is included in the stated 36-case cohort and excluded in a separate sensitivity analysis. The other 35 four-cell and all twelve five-cell instances were previously unevaluated.

The initial quality endpoint was unconditional optimum probability. Confirmation uses parameter seeds 11, 17, 23 and 31, giving 1,152 runs across two methods and three starts. Transfer uses seeds 41 and 53, with 576 original fixed-penalty runs including four-cell sampled training. Exact optima evaluate outcomes but are not the training loss. The reduced simulators retain $m!$ token states and m^n Row states: at six cells and eight sites these are 40,320 and 262,144. For fixed $n = 4, 5, 6$, $m!$ first exceeds m^n at $m = 7, 8, 10$, respectively.

The additional ring, workspace and coordinate comparisons were specified in commit 861fb30 after earlier results were known and before their implementation and execution. Ring adds 432 confirmation and 576 transfer runs with matched starts and seeds. Its pre-execution interpretation rule called a ring result competitive if it closed at least 75% of the positive calibrated four-cell mean gap, or matched token’s mean. This was a rule for interpreting the study, not an equivalence test. The extension, approximation and resource-budget controls were specified in commit e40a054 (specification SHA-256 prefix 8a98bc4d). Both sets are subsequent robustness analyses, not external preregistrations or new independent samples. The artifact preserves failures and separately specified execution repairs.

Penalty calibration uses three starts and seed 61 on eighteen development instances, giving 216 runs per graph. Selection is by problem size, with ties resolved by candidate order. The four-cell choice is reused for sampled training. The depth sensitivity sweep uses $p = 1, 2, 3, 4, 6$, plus any other depth selected by a resource budget. Its random-order token control uses independently seeded permutations of the same line edges for each layer and instance. Every distinct selected Row graph/depth is trained once per start and seed and reused across budgets that select it.

Bootstrap intervals use 20,000 within-family draws. This convention was adopted during initial confirmation analysis; later specifications fix seed 1144004 for the earlier controls and 1209505 for the additional endpoints. Paired differences are rounded to 10^{-12} for zero handling in the signed-rank tests. Those tests discard zero differences and

use Holm correction within the stated nine- or twelve-comparison families, separately by endpoint and test. The oracle comparison chooses among fixed and calibrated complete/ring baselines for each evaluation instance and endpoint; it is a sensitivity bound using information unavailable at tuning time.

A.1 Earlier synthesis and routing comparisons

The initial 100-case phase study used fixed mask order under Qiskit 2.4.1. L1 reduced CX relative to zero extension in all sixty cases with unused codes, with median reductions of 47.0%, 68.2% and 43.9% at six, nine and twelve sites; all forty power-of-two controls tied. The 60-case replication added beam and Gray synthesis, with Gray reductions of 27.6%, 53.9% and 21.3%. Its numerical comparisons also used angles 0.17, 0.371 and 1.03, taking angle medians within instances. The later 160-case extension comparison reran all methods under Qiskit 2.5.2 and used $\gamma = 0.371$ for numerical circuits. Results from different compiler versions or symbolic/numerical inputs are kept separate.

An early resource comparison bound token angles while leaving Row angles symbolic and retained redundant operations. Its apparent 15.3% ECR saving was withdrawn. A symmetric rerun used twenty routing seeds 101–120 on 36 confirmation and 32 earlier instances; the latter retained their rotating token schedule. Qiskit 2.4.1 native crashes led to a documented rerun of every routed case under 2.5.2. Token with zero extension and beam synthesis used more ECR gates on all 68 instance medians. On the 36 confirmation cases, changing to L1 with the same beam synthesis reduced token ECR by 18.2% and depth by 21.3%, but still exceeded complete-XY by 14.9% in ECR and 162.9% in depth. These counts use different phase synthesis from the main Gray comparison and are not combined with it.

The initial workspace study uses seeds 101–105; the main Gray/L1 comparison uses 211, 223, 227, 229 and 233. Software versions are Python 3.10.12, Qiskit 2.5.2, Runtime 0.46.1, NumPy 2.2.6 and SciPy 1.15.3; PyZX controls use version 0.10.3. Dependency locks accompany the artifact. The classical query-accounting correction reruns all 3,888 search records with instrumented objective calls, including initialization and repeated proposals; every search outcome is unchanged. Annealing temperature was selected on six earlier cases before transfer outcomes.

B Circuit Implementation Details

The following algorithms give the preparation, register swap and phase synthesis procedures used in the experiments. A mask is a bit set identifying the qubits in a Pauli- Z product. Historical *mask order* means increasing net index, then increasing integer mask; beam order retains separate net terms, whereas Gray synthesis combines equal supports across nets.

Algorithm 1 Token basis preparation

Input: injective assignment f , m valid sites, n real cells, $k = \lceil \log_2 m \rceil$. All qubits initially zero.

- (1) Set the label of real register u to $f(u)$.
- (2) Sort the unoccupied site labels and assign them, in order, to registers $n, \dots, m - 1$ representing distinguishable EMPTY tokens.
- (3) For each register, apply X to exactly its label's set bits. Keep all mixer ancillas zero. Output the resulting permutation basis state.

This uses at most mk single-qubit gates, no entangling gate, and depth at most one.

Algorithm 2 Clean-ancilla partial register swap $\exp(-i\beta S_{ab})$

Input: two k -qubit registers and one clean ancilla h .

- (1) Apply H_h ; for $i = 0, \dots, k - 1$, apply $\text{CSWAP}(h, a_i, b_i)$; apply H_h .
- (2) Apply $R_Z(2\beta)$ to h .
- (3) Apply H_h ; repeat the same k controlled swaps; apply H_h .

Output leaves $h = 0$ for every input superposition. Each Fredkin gate can be lowered without workspace as $\text{CX}(b_i, a_i)$, $\text{CCX}(h, a_i, b_i)$, $\text{CX}(b_i, a_i)$. Before lowering the Toffolis, this uses $4k$ CX, $2k$ Toffolis, four Hadamards and one variable rotation. A standard Toffoli implementation uses six CX, two Hadamards and seven $T/T^\dagger$ gates [33]. An explicit bound per partial swap is therefore $16k$ CX, $4k + 4$ Hadamards, $14k$ fixed $T/T^\dagger$ gates and one variable R_Z , before cancellation. The lowered component counts include all these gates in the declared basis. The shared control serializes this construction to $O(k)$ depth; no additional comparison or multi-control workspace is hidden in its width.

Let $A = H_h \text{CSWAP}_{ab} H_h$. On a SWAP eigenstate of eigenvalue $\sigma \in \{+1, -1\}$, A writes $(1 - \sigma)/2$ into h . The intervening rotation supplies phase $e^{-i\beta\sigma}$ and A uncomputes it. Thus the entire eigenspace-resolution cost is included in Algorithm 2; its $O(k)$ count establishes $O(pm \log m)$ primitive mixer work on a line. This upper bound precedes cancellation and hardware routing.

Algorithm 3 Valid-domain completion and parity synthesis

- (1) Form the m^2 valid distances, including equal-site pairs. Solve Eq. (5), or use an explicitly specified exact extension. Threshold coefficients at 10^{-10} and reject maximum valid residual $\geq 10^{-8}$.
- (2) For every net, map its pair-register mask to a global qubit support and multiply its coefficient by the net weight. Record the constant as global phase.
- (3) **Beam search:** start four paths with smallest (support size, mask, term index). Extend each path by up to ten unused terms, ranked by decreasing support overlap with its last term, then increasing symmetric difference, support size, mask and index. Keep the four paths with smallest explicit CX count after adjacent cancellation; retain stable ties. Repeat until complete. For each term compute parity onto the last bit in its ordered support list, rotate, then reverse its CX sequence, cancelling adjacent equal CX gates.
- (4) **Gray synthesis:** first sum coefficients with equal global supports. Group by highest bit, sort remaining control masks by inverse reflected-Gray rank, update only changed controls between rotations, and restore parity after finishing each target. Skip zero terms without approximating nonzero terms.

For adjacent supports A, B with the same target, appending B changes the score by $2(|B| - 1) - 2t$, where t is the common prefix length of their ordered control lists; use $t = 0$ for different targets. Each R_Z stops cancellation across earlier boundaries. Caching this score and static candidate rankings therefore preserves the archived beam decisions, including stable ties, while avoiding repeated partial-circuit reconstruction. The beam remains a bounded ordering heuristic, not an optimality certificate.

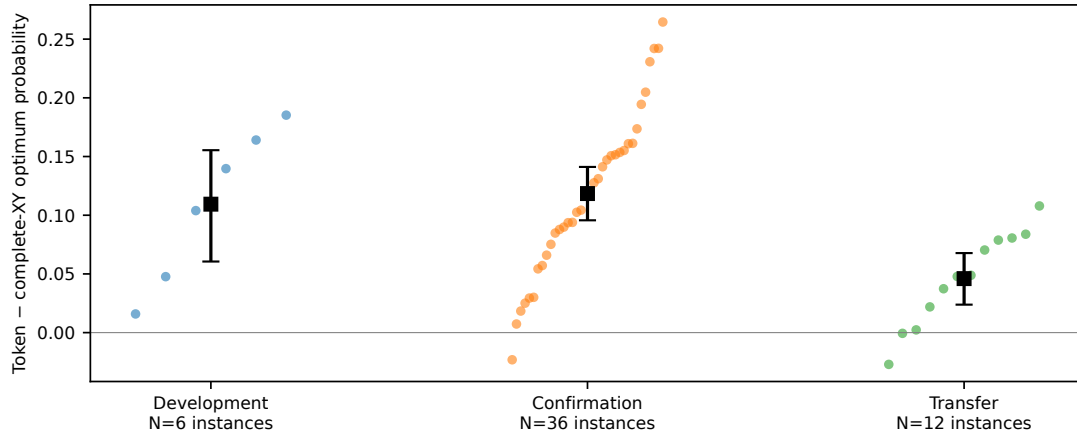


Fig. 3. Same-size cohort sensitivity. Points pair token with complete-XY within each four-cell/six-site instance; black squares and whiskers are cohort means and 20,000-draw instance-bootstrap 95% intervals. Cohorts are unpaired. Development uses two starts, two seeds, six original parameters and a 40-call cap; confirmation/transfer use three starts, four/two seeds and five free parameters. Their geometries and generators also differ.

C Quality Diagnostics

C.1 Cohort sensitivity

The initial confirmation gives token mean optimum probability 0.1558 versus 0.0373 for complete-XY on four-cell/six-site instances, with 35/36 positive paired differences. Excluding the smoke case gives 0.1526 versus 0.0349 and 34/35 positives. At five cells/seven sites the means are 0.0294 and 0.0016; only 6/12 poor-start differences are positive.

At four cells and six sites, the mean gap is 0.1094 in development, 0.1184 in confirmation and 0.0460 in transfer (Fig. 3). The gap therefore does not shrink monotonically from development onward. Because geometry, initialization and protocol also change, the transfer decrease cannot be attributed solely to selection bias.

C.2 Conditional and matched-shot endpoints

Conditioning on legal outputs changes the interpretation (Table 8). On confirmation, conditional optimum probability is 0.15579 for token, 0.08561 for complete-XY and 0.09245 for ring; ring closes 9.8% of this conditional gap. On calibrated four-cell analytic transfer, the corresponding values are 0.05713, 0.02747 and 0.02195. The token-minus-ring conditional difference is 0.03518 with paired interval [0.00353, 0.06412], but its post hoc Wilcoxon test does not survive Holm correction across the nine cohort/training conditions ($p_{\text{Holm}} = 0.448$). Confirmation does ($p_{\text{Holm}} = 0.000827$). These descriptive robustness tests temper a claim based on the original optimum-probability endpoint alone.

Matched-shot best cost gives another view. With 128 final shots, calibrated four-cell analytic token and ring means are 23.553 and 24.151: a paired difference of -0.597 with interval $[-1.137, -0.169]$, favoring token. The corresponding sampled-training difference is $+0.126$ with interval $[-0.459, 0.907]$. Thus the analytic conditional and best-shot results do not establish a sampled-training advantage. Every shot, including an illegal Row output, is charged; the known initial placement is the fallback.

Table 8. Legality and conditional quality at matched $p = 3$. L is legal probability; Q is the run-wise ratio of optimum to legal probability, averaged over starts/seeds and then instances. Subscripts T, C, R denote token, complete-XY and ring-XY; $L_T = 1$ throughout. Confirmation has 36 instances; each transfer condition has 12. Fixed uses $\lambda = 5$; calibrated penalties are selected independently for each Row graph. Closure is $(Q_R - Q_C)/(Q_T - Q_C)$ and is undefined when token does not exceed complete-XY. Shots denotes 2,048-shot training; endpoints use the final ideal distribution. Postselection still incurs the discarded measurement shots.

Cohort	Size	Train	L_C	L_R	Q_T	Q_C	Q_R	Closure
Confirm.	4	exact	0.375	0.447	0.156	0.0856	0.0925	9.8%
Fixed	4	exact	0.336	0.390	0.0571	0.0228	0.027	12.0%
Fixed	4	shots	0.327	0.376	0.0264	0.0144	0.0241	81.4%
Fixed	5	exact	0.252	0.368	0.000447	0.00281	8.73e-05	–
Fixed	6	exact	0.137	0.232	0.000188	0.00269	2.48e-05	–
Calibrated	4	exact	0.396	0.412	0.0571	0.0275	0.0219	-18.6%
Calibrated	4	shots	0.379	0.396	0.0264	0.0296	0.0231	–
Calibrated	5	exact	0.269	0.477	0.000447	0.00328	9.85e-06	–
Calibrated	6	exact	0.242	0.348	0.000188	0.000122	3.59e-05	-131.2%

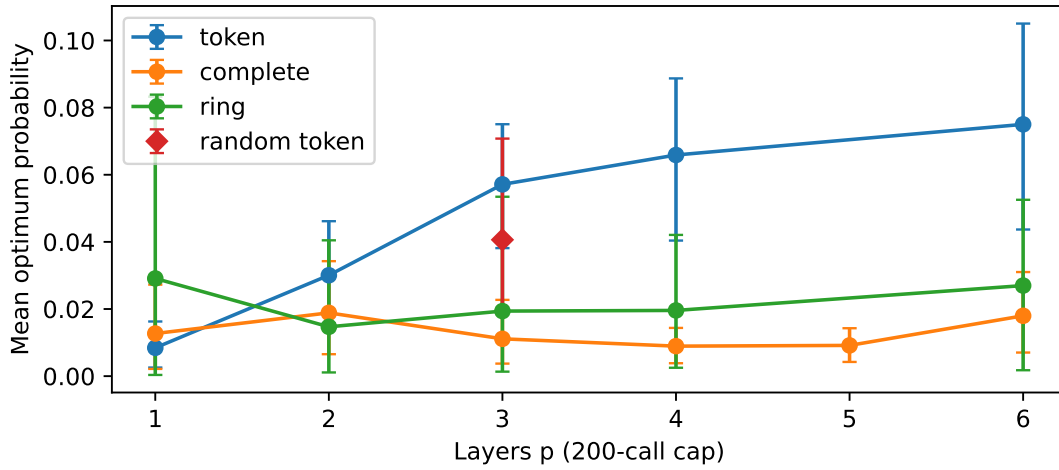


Fig. 4. Depth sensitivity under fixed $\lambda = 5$, three starts, two seeds and the same 200-call COBYLA cap. Points are twelve-instance mean optimum probabilities; whiskers are family-stratified bootstrap 95% intervals. Additional selected Row depths are included; the random token schedule is a separately seeded $p = 3$ control. Different depths are different ansatzes.

The evaluation oracle discussed in Section 8 also reduces token’s unconditional optimum-probability difference to 0.02605, with interval $[-0.00014, 0.05396]$. Full paired summaries, including medians, win/tie/loss counts and corrected tests, are available in the artifact.

C.3 Depth and schedule sensitivity

The depth sweep (Fig. 4) is not monotone for the Row mixers: ring’s $p = 1$ mean is 0.02911, slightly above its $p = 6$ mean, whereas token increases from 0.00844 at $p = 1$ to 0.07501 at $p = 6$. A separately seeded random token schedule at $p = 3$ gives 0.04063, below the declared schedule’s 0.05713. This descriptive schedule control does not isolate a causal effect of EMPTY prioritization.

Table 9. Numeric synthesis controls on the replication cohort, twelve cases per site count. Entries are paired median CX percentage changes relative to numeric Gray for the same extension. Negative means fewer CX than Gray. Other extensions and power-of-two controls are retained in the artifact.

Extension	m	Generic / Gray (%)	PyZX / Gray (%)
L1	6	+32.2	+125.3
L1	9	+107.0	+130.4
L1	12	+30.5	+167.1
virtual	6	+8.4	+203.8
virtual	9	+4.6	+226.1
virtual	12	+4.6	+245.4

D Additional Synthesis Diagnostics

The numeric controls hold $\gamma = .371$ fixed and compare each exact extension with the same fixed angle (Table 9). Extracted PyZX circuits are checked against their parsed inputs; the artifact separately records OpenQASM angle-conversion bounds, so a symbolic exactness claim is not inferred from floating-point parsing. Extension solve times, synthesis times, weighted norms, support sizes, one-qubit rotations and depth accompany every case. Timings describe these implementations and allocations, not asymptotic performance or a cross-machine speed benchmark. The execution record identifies identical-input PyZX cache hits; mixed cache timings are not an uncached compiler-speed comparison.

Independent reconstruction checks all 1,760 stored coefficient vectors. The maximum valid-distance residual for exact extensions is 5.69×10^{-14} . Across 800 numeric phase inputs, QASM angle conversion has a maximum triangle-inequality operator-error bound of 4.76×10^{-8} up to global phase. These two tolerances concern different objects. PyZX’s full-reduction and extraction pipeline uses more CX than numeric Gray in every tested input; for replication L1 cases its paired median excess is 125.3%, 130.4% and 167.1% at six, nine and twelve sites. This result applies to that PyZX reduction and extraction procedure.

Table 10 summarizes approximate synthesis. At $\eta = .001$, approximate truncation leaves the replication median CX unchanged at all three non-power-of-two sizes. At $\eta = .05$, L1 saves a further 6.3%, 15.3% and 16.3%, and virtual completion saves 4.5%, 11.9% and 10.0%, relative to each exact input. The corresponding maximum L1 valid residuals are 1.375, 1.78125 and 2.84375 distance units. This additional saving buys a bounded objective perturbation; it is not evidence of unchanged optimization quality.

Figure 5 shows how completion changes the distribution of Pauli weights. Equal CX counts need not imply equal Walsh spectra. With zero extension and Gray synthesis, the nine- and twelve-site replication cases both use four site bits and yield 1,214 CX, but their median aggregated rotation counts are 1,170 and 1,185, with twelve and ten distinct support sets across their respective twelve cases. At sixteen sites the median is 1,333 CX; the same binary width does not fix the active parity network.

E Local Placement Integration with OpenROAD

A separate integration study inserts local assignments into the OpenROAD Flow Scripts (ORFS, commit `be0dca0b1fd4`) using Nangate45 and a recorded container image. It covers GCD, AES, Ibex, Mempool Group, SweRV and TinyRocket. For each design, a deterministic extractor produces 120 candidate four-cell/six-site windows with fixed external net anchors and selects four cell-disjoint windows from each connectivity tertile by a seeded hash ordering. The artifact

Table 10. Greedy approximate Gray controls on twelve replication cases per site count. CX changes are paired medians relative to the same exact extension. The maximum observed valid-distance residual is reported alongside the allowed relative tolerance η ; they are different quantities. All three declared tolerances, power-of-two cases and assignment-error bounds are in the artifact.

Extension	m	η	CX change (%)	Max. valid residual
L1	6	0.001	+0.0	0
L1	6	0.050	-6.3	1.38
virtual	6	0.001	+0.0	0
virtual	6	0.050	-4.5	1.44
L1	9	0.001	+0.0	0.0312
L1	9	0.050	-15.3	1.78
virtual	9	0.001	+0.0	0.0312
virtual	9	0.050	-11.9	1.83
L1	12	0.001	+0.0	0.0312
L1	12	0.050	-16.3	2.84
virtual	12	0.001	+0.0	0.0469
virtual	12	0.050	-10.0	2.72

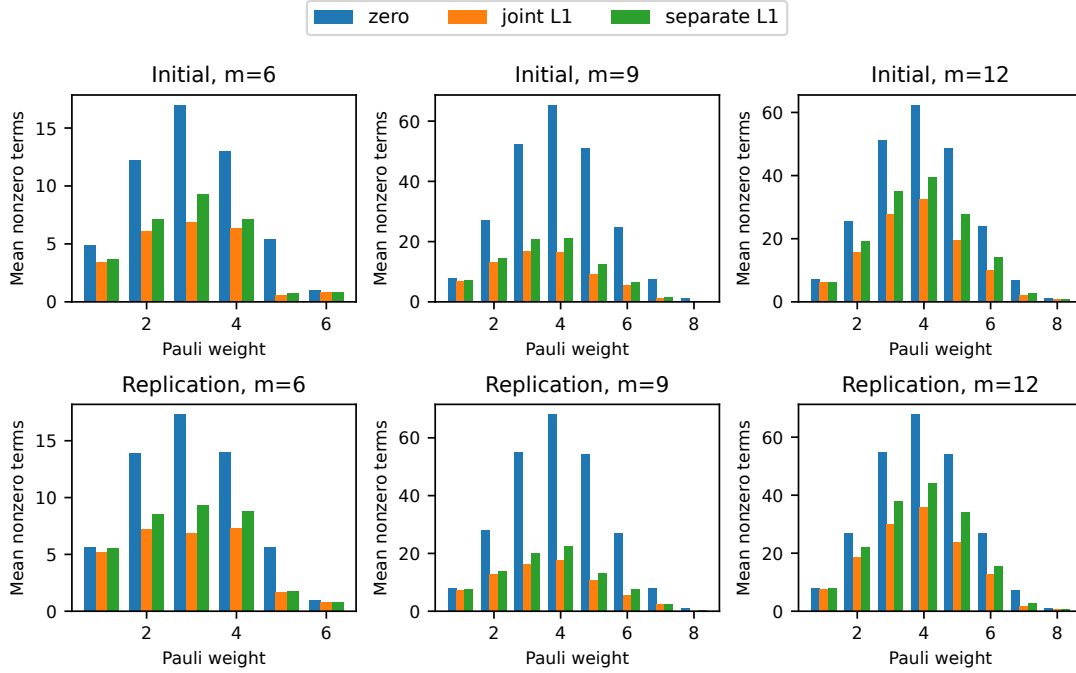


Fig. 5. Numbers of two-register Walsh terms by Pauli weight (mask popcount) after the 10^{-10} numerical threshold. Bars average 20 initial-cohort or 12 replication-cohort cases at each site count. These are coefficient supports before net aggregation and synthesis; they are not final CX counts. Zero, joint L1 and coordinate-separated L1 use the retained coordinate-control vectors.

specifies the exact selection rule. The resulting 72 windows are nested within six designs; 24 are initially optimal. This study uses a ring/reversed token schedule, an improvement-probability objective, four optimizer seeds and 4,096 final measurements per optimizer seed, rather than the synthetic protocol in the main text.

There are only 360 legal assignments per window. With g optimal assignments, uniform feasible sampling finds one in B draws with probability $1 - (1 - g/360)^B$, which approaches one at the study's readout budget. QAOA finds an optimum in 71 of 72 windows. Greedy search finds 69 with 32 neighbor evaluations (33 objective queries including the known initial cost), and both recorded classical searches find all 72 by 360 queries per seed. Among the 48 windows not initially optimal, these counts are 47 for QAOA and 45 for short greedy search; both classical methods again solve all 48 by 360 queries. QAOA training incurs additional work.

For each window, the downstream comparison inserts the selected QAOA assignment and assignments from exact search, greedy search and simulated annealing into the design flow. The resulting 288 assignments give the completion counts in Table 7. These checks establish that the selected local placements can be used by subsequent flow stages; zero routing overflow does not by itself establish improved timing or routed wirelength.

F Notation

Table 11 collects the principal symbols. Site coordinates x_s and binary assignment variables $x_{u,s}$ are distinct; the layer index ℓ and the coordinate bit width ℓ are local to different constructions.

Table 11. Principal notation, with the section in which each meaning is introduced.

Symbol	Meaning	Section
n, m	Numbers of real cells and valid sites	3
$r_s = (x_s, y_s), f$	Site coordinates and an injective cell-to-site assignment	3
$E, w, C(f), d(a, b)$	Weighted nets, a net weight, placement cost and Manhattan distance	3
$x_{u,s}, \lambda$	One-hot assignment variable and penalty parameter	3
k, p	Bits per site register and QAOA layers	3
$\beta_\ell, \gamma_\ell, \theta$	Mixer angle, phase angle and parameter vector; ℓ indexes layers	3
z, j, W, Z_j	Encoded pair, bit mask, Walsh matrix and Pauli-Z product	4
$c_j, \tilde{d}, V, \epsilon$	Walsh coefficient, completed distance, valid encoded pairs and maximum valid-pair residual	4
ℓ, a', b', A, B	Coordinate bit width, lower-bit integers and high-bit signs in the recurrence	4.1
T_ℓ, k_x, k_y	Nonzero coefficient count and coordinate bit widths	4.1
D, η	Valid-site diameter and relative truncation tolerance	4.3
$B, d_{\max}$	Penalty bound $Dd_{\max}/2$ and maximum weighted net degree	5.3
F	Fraction of distance-table entries not fixed by valid labels	6
L, Q	Legal probability and optimum probability conditional on legality	C
g, B	Number of optimal assignments and sampling budget in the OpenROAD null model	E

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